Let us take up the case where $x \neq 0,\ y \neq 0$.

Dividing both sides of each equation $by\ xy$,

$$\frac{8x}{xy} - \frac{3y}{xy} = \frac{5xy}{xy} \qquad \text{we get,} \qquad \frac{8}{y} - \frac{3}{x} = 5 \quad ...(3)$$

$$\frac{6x}{xy} - \frac{5y}{xy} = \frac{-2xy}{xy} \qquad \frac{6}{y} - \frac{5}{x} = -2 \quad ...(4)$$

Let $a = \frac{1}{x},\ b = \frac{1}{y}$.

(3)&(4) respectively become, $8b - 3a = 5$ ...(5)

$$6b - 5a = -2 \quad ...(6)$$

which are linear equations in a and b.

To eliminate a, we have, $(5) \times 5 \Rightarrow \quad 40b - 15a = 25$ .....(7)

$(6) \times 3 \Rightarrow \quad 18b - 15a = -6$ .....(8)

Now proceed as in the previous example to get the solution $\left(\frac{11}{23}, \frac{22}{31}\right)$.

Thus, the system have two solutions $\left(\frac{11}{23}, \frac{22}{31}\right)$ and $(0,0)$.

Exercise 3.12

1. Solve by the method of elimination

   (i) $2x - y = 3;\quad 3x + y = 7$ (ii) $x - y = 5;\quad 3x + 2y = 25$

   (iii) $\frac{x}{10} + \frac{y}{5} = 14;\quad \frac{x}{8} + \frac{y}{6} = 15$ (iv) $3(2x + y) = 7xy;\quad 3(x + 3y) = 11xy$

   (v) $\frac{4}{x} + 5y = 7;\ \frac{3}{x} + 4y = 5$ (vi) $13x + 11y = 70;\ 11x + 13y = 74$

2. The monthly income of A and B are in the ratio 3:4 and their monthly expenditures are in the ratio 5:7. If each saves ₹ 5,000 per month, find the monthly income of each.

3. Five years ago, a man was seven times as old as his son, while five year hence, the man will be four times as old as his son. Find their present age.

Solving by Cross Multiplication Method

The substitution and elimination methods involves many arithmetic operations, whereas the cross multiplication method utilize the coefficients effectively, which simplifies the procedure to get the solution. This method of cross multiplication is so called because we draw cross ways between the numbers in the denominators and cross multiply the coefficients along the arrows ahead. Now let us discuss this method as follows:

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